UNIT – II

Method Study • Motion Economy • Work Measurement • Work Sampling

Syllabus (as given)

Method Study: definition, objective and scope, activity recording and tools – Outline Process Chart, Flow Process Chart, Flow Diagram, String Diagram, Travel Chart, Multiple Activity Chart, Two-Handed Process Chart. Principles of Motion Economy: classification of movements, micromotion study, Therbligs, SIMO chart. Special charts: Cyclegraph and Chronocyclegraph – development, definition and installation of the improved method.

Work Measurement: definition, objectives, techniques. Work sampling – need, confidence levels, sample size determination, conducting a study with problems.

1. Method Study – Definition, Objective, Scope

Definition (ILO)

Method study is the systematic **recording** and **critical examination** of existing and proposed ways of doing work, as a means of developing and applying easier and more effective methods and reducing costs.

Objectives

- Improvement of processes and procedures; improvement of factory, plant and work-place layout.
- Economy of human effort and reduction of unnecessary fatigue.
- Economy in the use of materials, machines and manpower; improved material handling.
- Development of a better physical working environment; **standardisation** of the improved method.
- Simplifying work so that the same output is obtained with less effort and cost, and better safety.

Scope

- Applicable to every activity involving human work – shop floor, offices, stores, maintenance, hospital, banks, construction, etc.

- Studies: the process as a whole (flow of material / men), a single operation, a work-place, or even a single hand motion (micromotion).

Basic procedure (7 steps)

1. **SELECT** the job. Factors: **economic** (high cost, bottleneck, large volume), **technical** (quality problems, rejects, difficult work), **human** (fatigue, accidents, union complaints, boring work).
2. **RECORD** all relevant facts of the present method by charts, diagrams, film. (Direct observation – accurate, complete, concise.)
3. **EXAMINE** the facts critically using the questioning technique.
4. **DEVELOP** the most practical, economic and effective method; consult workers; compare alternatives with costs.
5. **DEFINE** the new method – written standard practice, drawings, charts.
6. **INSTALL** the new method as standard practice – get approval, train, trial run, change over.
7. **MAINTAIN** the standard by periodic review; prevent drifting back to the old method.

Critical examination – the questioning technique

Aspect	Primary question	Secondary question	Aim
Purpose	What is done? Why is it done?	What else could be done? What should be done?	ELIMINATE
Place	Where is it done? Why there?	Where else could it be done? Where should it be done?	COMBINE / REARRANGE
Sequence	When is it done? Why then?	When else could it be done? When should it be done?	REARRANGE
Person	Who does it? Why that person?	Who else could do it? Who should do it?	COMBINE / REARRANGE
Means	How is it done? Why that way?	How else could it be done? How should it be done?	SIMPLIFY

Note: ECRS principle: Eliminate – Combine – Rearrange – Simplify (in this order of priority). Develop the new method using the answers.

2. Activity Recording and Tools

- Recording must be **accurate, complete and concise** – it is the base of the examination. Never trust memory.

Group	Tool	Used to show
Charts – process	Outline process chart, Flow process chart (man / material / equipment type)	Sequence of events in a process, with symbols
Charts – activity (time-scaled)	Multiple activity chart, Two-handed process chart, SIMO chart	Simultaneous activities of men / machines / hands against time
Diagrams (scaled layout)	Flow diagram, String diagram, Travel chart (From–To chart)	Paths of movement on a layout; frequency of movement
Photographic	Cine film, micromotion, memo-motion, time-lapse, video; Cyclegraph, Chronocyclegraph	Fine motions, patterns of movement

Process chart symbols (ASME)

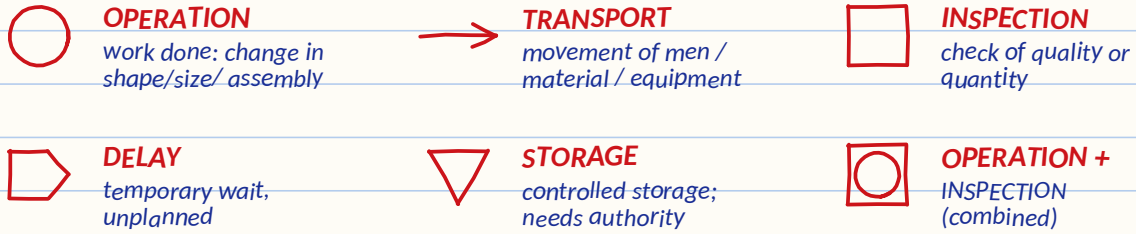


Fig 2.1 Standard (ASME) symbols; size of circle ≈ 10 mm

- Operation = productive step (value-adding). Transport, Delay, Storage = non-productive (try to eliminate). Inspection = non-value-adding but necessary.

(a) Outline Process Chart (OPC)

- Gives an **overall picture** – shows only the **operations (O)** and **inspections (I)** in sequence, and the points where materials enter the process. Also called an "operation process chart".
- Main flow line vertical on the right; material lines enter horizontally; sub-assemblies enter from left. Time can be written on the right.

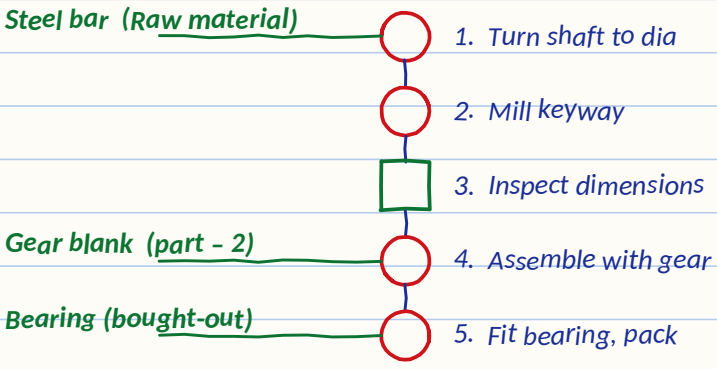


Fig 2.2 Outline process chart for a shaft–gear assembly

(b) Flow Process Chart (FPC)

- More detailed than OPC – shows **all five activities:** operation, transport, inspection, delay, storage; with distance (m) and time (min).
- Three types: **Man type** (what the worker does), **Material type** (what happens to the material), **Equipment type** (how the equipment is used).
- Used for analysing the sequence, to expose transport, delays and storage – then apply ECRS. Compare "Present method" and "Proposed method" with a summary table.

No.	Description of activity	Dist(m)	Time	O T I D S
1	Raw bar in store			
2	Move bar to saw by trolley	25	2.0	
3	Wait for saw		6.0	
4	Cut to length		3.0	
5	Carry to lathe	15	1.5	
6	Turn and drill		9.0	
7	Inspect dia and length		1.0	
8	Move to finishing	20	1.5	
9	Store in bin			

Fig 2.3 Flow process chart – material type (present method)

Summary	Operation	Transport	Inspect	Delay	Storage
No. of events	2	3	1	1	2
Time (min)	12.0	5.0	1.0	6.0	–
Distance (m)	–	60	–	–	–

(c) Flow Diagram

- A **scale layout** of the floor / building on which the path of movement (men / material) is drawn, numbered to match the flow process chart. Shows back-tracking, crossing, long distances, congestion.
- Direction shown by arrows; symbols O T I D S marked at the points of occurrence; different colours for different materials.

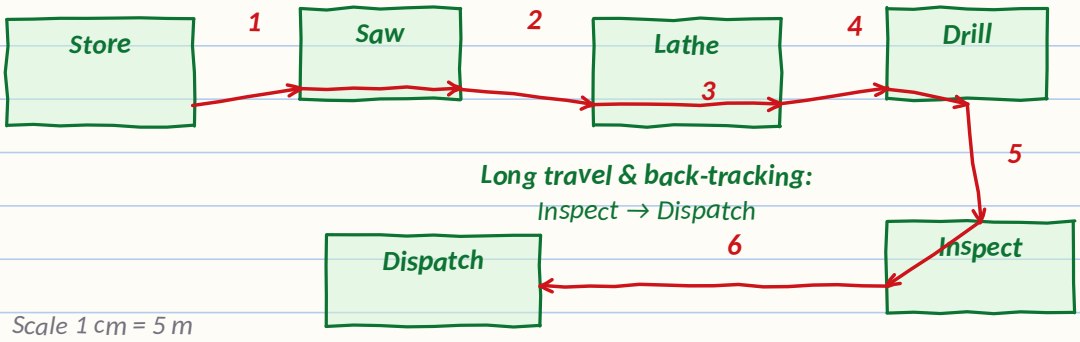


Fig 2.4 Flow diagram of present method (numbers = process chart steps)

(d) String Diagram

- A scale layout with **pins at each work-station;** a continuous **thread** is wound round the pins in the order of the movements. Length of thread used × scale = total distance travelled.
- Used for: movement of a worker / material over a shop floor, repetitive movements, crowded areas – to design a better layout (shorter strings), compare layouts.
- Distance = (string length used) × (scale factor). Heavier string concentration = **more frequent movement** → bring those stations closer.

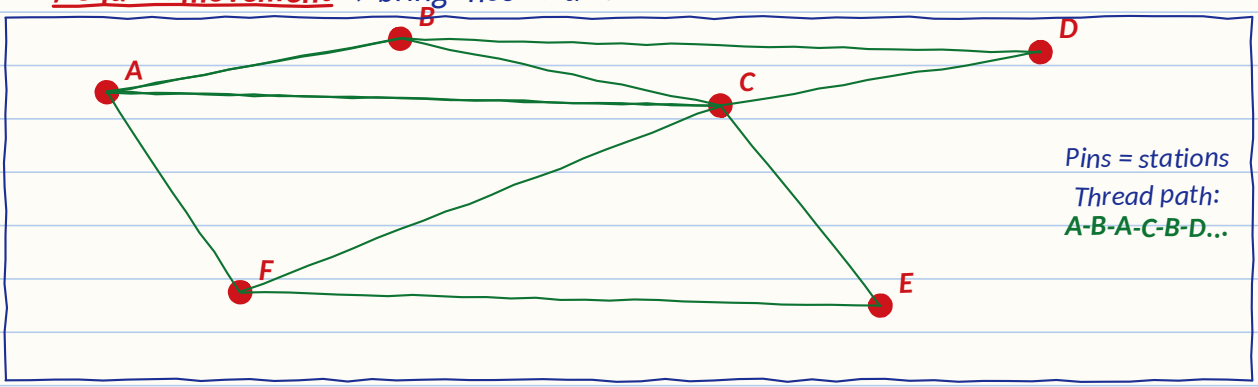


Fig 2.5 String diagram of a work-area

Flow diagram	String diagram
Shows direction & path of flow, single trip	Shows frequency and total travel (repeated movement)
Drawn with lines & arrows; easy and quick	Needs board, pins and thread; more time, more accurate for distance
Good for process flow of material	Good for worker movement, layout of tool crib, stores, offices

(e) Travel Chart (From–To chart)

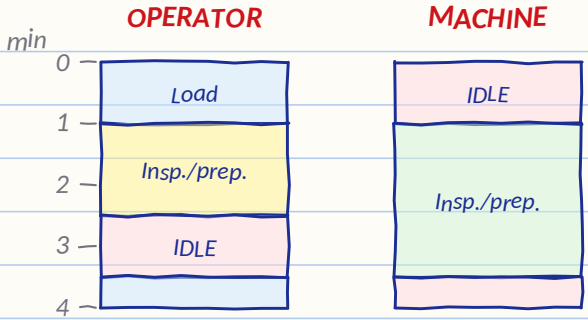
- A tabular record of the **number of movements** (loads or trips) between work-stations / departments in a given period. Used for layout planning – place stations with high travel close to each other.
- Rows = FROM, columns = TO. Frequency × distance gives the material-handling load; the sum of (trips × distance) is to be minimised.

From \ To	Store	Saw	Lathe	Drill	Inspect
Store	–	40	10	–	–
Saw	–	–	35	15	–
Lathe	–	–	–	30	20
Drill	–	–	–	–	45
Inspect	5	–	–	–	–

Note: Highest flows: Drill→Inspect (45), Store→Saw (40), Saw→Lathe (35) – place these pairs adjacent.

(f) Multiple Activity Chart (Man–Machine chart)

- Records the activities of **several subjects (men / machines)** on a common time scale – shows the idle time of each and how activities are inter-related. Used for man–machine and gang (team) work.
- Used to: find idle (waiting) time, improve utilisation, decide the **number of machines one operator can attend,** balance team work.



Cycle time = 4.0 min
Operator utilisation = 3.0/4.0 = 75 %
Machine utilisation = 2.5/4.0 = 62.5 %
Output = 60/4 = 15 parts / hour

Fig 2.6 Multiple activity chart (one man – one machine)

Number of machines per operator (synchronous servicing)

n = (l + m) / (l + w) l = operator loading / unloading time per machine, m = machine running time, w = walking time between machines.

Example: l = 1.5, m = 2.5, w = 0.5 → n = 4.0 / 2.0 = 2. Choose n = 2 (or compute cost per piece for n = 2 and 3 and pick the lower).

(g) Two-Handed Process Chart (Left-hand / Right-hand chart)

- Records the **movements of both hands** of a worker in relation to one another on a time scale – used for repetitive short-cycle bench / assembly work to expose idle hand, holding, unbalanced work.
- Symbols: O (operation – grasp, position, use, release), T (transport – reach, move), D (delay – idle/waiting), (hold). Hold is shown with the inverted triangle.

Left hand	Sym	Sym	Right hand
Reach for bolt	T	T	Reach for nut
Grasp bolt	O	O	Grasp nut
Move bolt to centre	T	T	Move nut to centre
Hold bolt	S (hold)	O	Screw nut on bolt
Hold bolt	S (hold)	O	Tighten with spanner
Place assembly in bin	T	D	Wait / idle

Note: Improvement: use a fixture to HOLD the bolt (frees LH), use drop-delivery bins; both hands then work together.

3. Principles of Motion Economy

- Introduction: Frank & Lillian **Gilbreth** developed motion study; Barnes later formulated the **principles of motion economy** – rules to reduce fatigue and increase output by eliminating useless motions. They are a check-list while developing the new method.

Classification of movements (Barnes / Gilbreth)

Class	Body members that move	Example
1	Fingers only	Pressing a push button, typing
2	Fingers + wrist	Turning a small knob, using a screwdriver
3	Fingers + wrist + forearm (elbow pivot)	Reaching for a tool on the bench
4	Fingers + wrist + forearm + upper arm (shoulder pivot)	Reaching across the bench
5	Class 4 + trunk (body movement)	Bending / twisting to reach a bin on the floor

Note: Rule: do the work with the lowest possible class of movement (class 1 or 2) – it needs least time and fatigue.

- Other classifications: **Fundamental hand motions** (reach, move, grasp, position, release, etc. = therbligs); motions can also be: ballistic vs controlled, symmetrical, continuous curved vs straight-line with sharp turns.

Barnes' principles – three groups

(I) Use of the human body

- Both hands should begin and complete their motions at the same time; and should not be idle at the same time except during rest.
- Motions of the arms should be **symmetrical and in opposite directions** and made simultaneously.
- Hand motions should be confined to the **lowest classification** that can do the work satisfactorily.
- Momentum should be employed to assist the worker; if it must be overcome by muscular effort, reduce it to a minimum.
- Smooth continuous curved motions are preferable to straight-line motions involving sudden sharp changes of direction.
- Ballistic (free swing) movements are faster, easier and more accurate than restricted (controlled) movements.
- Rhythm is essential to smooth, automatic performance; work should be arranged to permit an easy natural rhythm.

(II) Arrangement of the work-place

- There should be a **definite and fixed place** for all tools and materials.
- Tools, materials and controls should be **pre-positioned** and located close in front of the worker – within the normal working area.
- Gravity-feed bins / containers and drop-delivery chutes should be used to deliver material close to the point of use.
- Materials and tools should be arranged to permit the best sequence of motions.
- Provide adequate illumination; chair and work height should permit alternate sitting & standing; chair of proper design for good posture.
- Eye-fixation should be reduced: visual displays and parts in the normal vision field.

(III) Design of tools and equipment

- Use a jig, fixture or foot-operated device to **relieve the hands of holding** work – hands then do useful work.
- Two or more tools should be combined wherever possible (e.g. combination pliers).
- Tools and materials should be pre-positioned whenever possible (ready for the next motion).
- Where each finger performs a specific movement (typing), the load should be distributed according to their capability.
- Handles (e.g. crank, screw-driver) should be designed to give as much surface contact as possible with the hand; levers, cross-bars and hand wheels should be positioned for maximum mechanical advantage.



Normal area (Barnes): arc swung by forearm pivoting at elbow → used for frequent work

Max. area: arm swung from shoulder → used for occasional work only

Fig 2.7 Normal and maximum working areas in the horizontal plane

4. Micromotion Study, Therbligs and SIMO Chart

- **Micromotion study** (Gilbreth, 1912): detailed study of the very short cycle repetitive work by **filming the operation** (cine camera / video) with a **microchronometer** (a clock reading 1/2000 min) in the field of view – then analysing frame by frame.
- Uses: to analyse and improve fine hand motions of highly repetitive jobs of long run; to train operators; to find the best motion sequence; as a research tool for developing standard data and PMTS.
- Procedure: (1) select job & prepare work-place, (2) film at standard speed (e.g. 1000 frames / min) with the clock, (3) analyse film – break into therbligs, (4) record on SIMO chart, (5) critically examine, (6) develop & film new method, (7) install.
- Limitation: high cost of cine camera / film, time to analyse – justified only for high-volume work. Memo-motion (time-lapse filming at 1 frame per 1/2 – 2 s) is a cheaper tool for long-cycle & multi-person activity.

Therbligs (Gilbreth spelt backwards – "Gilbreth" → "Therblig")

- The basic **elements of hand motion:** 17 (some texts list 18 including Hold / Find). Each has a symbol, letter code and a colour.

Therblig	Code	Description	Type
Search	Sh	Eyes / hand groping to find an object	Ineffective
Select	St	Choosing one article among several	Ineffective
Grasp	G	Closing fingers to hold an object	Effective
Transport empty (Reach)	TE	Moving the empty hand towards an object	Effective
Transport loaded (Move)	TL	Moving an object with the hand	Effective
Hold	H	Retaining the object in a fixed position	Ineffective
Release load	RL	Letting go of the object	Effective
Position	P	Orienting the object during use	Effective / semi
Pre-position	PP	Placing the object ready for the next operation	Effective
Inspect	I	Examining the object for quality	Ineffective / necessary
Assemble	A	Joining two parts together	Effective (productive)
Disassemble	DA	Separating parts	Effective (productive)
Use	U	Using a tool / device for the intended purpose	Effective (productive)
Unavoidable delay	UD	Delay beyond the worker's control	Ineffective
Avoidable delay	AD	Delay within the worker's control	Ineffective
Plan	Pn	Mental process – deciding the next action	Ineffective
Rest (to overcome fatigue)	R	Pause needed to recover from fatigue	Ineffective (needed)
Find	F	Eyes locating an object (end of Search)	Ineffective

Note: Aim: eliminate / shorten ineffective therbligs (Sh, St, H, P, PP, I, Pn, AD, UD) using the principles of motion economy; keep the effective ones (TE, G, TL, A, U, RL) short.

SIMO Chart (Simultaneous Motion Cycle Chart)

- A **two-handed chart in therbligs,** drawn on a common time scale (in **winks:** 1 wink = 1/2000 min = 0.0005 min) from the micromotion film. Shows both hands' therbligs side by side with their durations (in colour codes) – for exact analysis of short repetitive cycles.

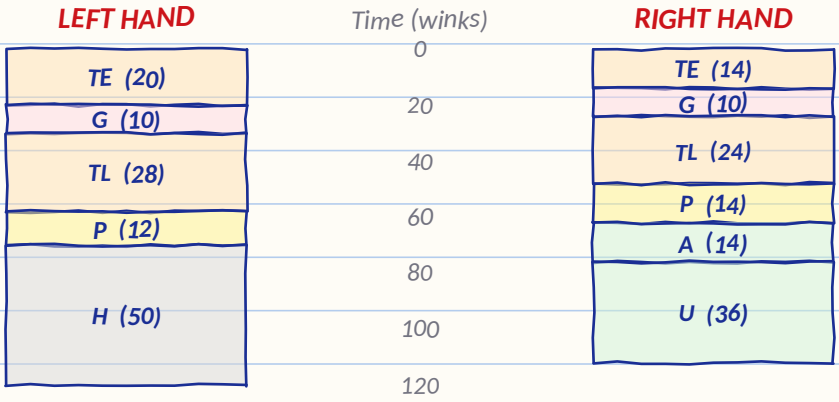
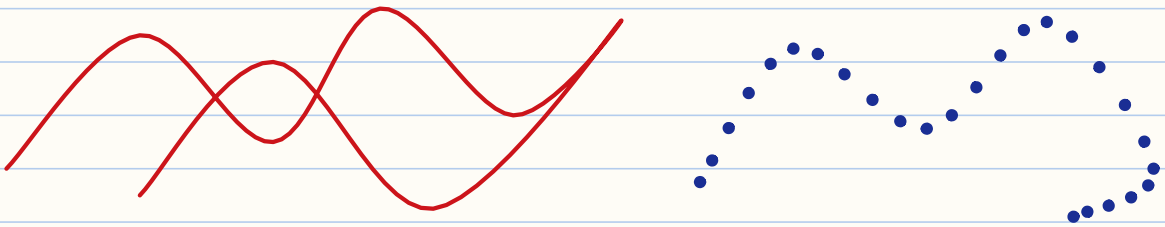


Fig 2.8 SIMO chart (schematic): note H (hold) of LH during RH working → provide a fixture

5. Special Charts – Cyclegraph and Chronocyclegraph

- **Cyclegraph** (Gilbreth, 1917): a record of the **path of a motion** – a small light bulb is fixed to the worker's finger / hand and a still camera takes a long-exposure (time) photograph in a dark room → a continuous bright line shows the path of motion (design of simpler paths, training).
- **Chronocyclegraph:** same but the light is **interrupted at a known rate** (by a rotating shutter or pulsating lamp) → the path appears as a series of dots / dashes (pear-shaped). Spacing of dots tells **speed, acceleration, deceleration, time** of the motion – dots close = slow; far = fast.



Cyclegraph (continuous trace) Chronocyclegraph (dots = time)

Fig 2.9 Cyclegraph and Chronocyclegraph (schematic)

- Applications: to find the ideal motion path, to compare operators, to design work-place layout and training; to study the motions of skilled operators, also in sports and rehabilitation.
- Memo-motion: time-lapse film, used for gang / multi-man activity; micromotion: high-speed fine analysis.

Development, Definition and Installation of the improved method

- **DEVELOP:** apply ECRS and motion economy; prepare alternatives; compare on cost, safety, quality, flexibility; test (pilot trial); select the best.
- **DEFINE:** write a standard-practice sheet (operation, tools, work-place sketch, sequence of motions, standard time); get the approval of management.
- **INSTALL:** (i) get management approval, (ii) win acceptance of supervisors & workers, (iii) train operators step by step, (iv) arrange tools, material, layout, (v) trial run and follow-up, (vi) start full production.
- **MAINTAIN:** audit regularly; settle complaints; update the standard when changes occur; reward people for suggestions.

6. Work Measurement

Definition (ILO)

Work measurement is the application of techniques designed to establish the time for a **qualified worker** to carry out a specified job at a **defined level of performance.**

Objectives / uses

- To fix standard time for output planning, production scheduling, machine loading, delivery dates.
- To estimate manpower and equipment; to balance assembly lines and teams; cost estimation and standard costs.
- To compare alternative methods (after method study); to measure ineffective time and efficiency; to set incentive wages and control labour cost.

Techniques of work measurement

Technique	Brief idea	Best for
Time study (stop-watch)	Direct timing of elements with performance rating	Repetitive short-cycle jobs
Synthesis (standard data)	Build time from stored element times of earlier studies	Jobs with common elements, new jobs
PMTS (MTM, WF, MOST)	Standard times for basic motions from tables	Planning a method before it exists
Work sampling / activity sampling	Random observations of activity → % of time	Long cycle, non-repetitive, many workers
Analytical estimating	Estimate element times by an experienced person	Maintenance, long jobs

Technique	Brief idea	Best for
Estimating / historical data	Past records, rough judgement	Quick, low-accuracy

7. Work Sampling (Activity Sampling / Ratio-Delay Study)

- Developed by L.H.C. **Tippett** (UK textile mills, 1930s). A large number of **instantaneous observations** taken at random intervals of a group of machines / workers; each observation records the activity (working / idle / type of delay) – the fraction of observations ≈ fraction of time.

Need / when to use

- Long-cycle, irregular, non-repetitive or many-person (gang) work; determination of **machine / worker utilisation** and the delay %.
- To find allowances (fatigue, personal, delay); to set the standard time for non-repetitive jobs; to estimate staff requirement (e.g. in offices, maintenance).
- Advantages: observer need not stay continuously; no stop watch; operator not nervous; one observer can study many subjects; can be interrupted & resumed; cheap. Disadvantage: does not give detailed element time; not suitable for short-cycle jobs; observer must be unbiased.

Theory – confidence level & sample size

- Based on the **binomial distribution** (approx. normal for large n). If p = fraction of observations in the state (e.g. idle), n = number of observations, the std. deviation of p is $\sigma_p = \sqrt{p(1-p)/n}$.

Confidence level	Z value (± z σ)	Meaning
68.27 %	1	true p within p ± 1σ
90 %	1.645	p ± 1.645 σ
95 % (most used)	1.96 ≈ 2	p ± 2σ (practical value)
99 %	2.576	p ± 2.58 σ
99.73 %	3	p ± 3σ

Sample size formulae (95 % confidence, z = 2)

Absolute error: $n = 4p(1-p)/L^2$ where L = absolute accuracy (e.g. ±0.05 for ±5 %)

Relative error: $n = 4(1-p)/(S^2 p)$ where S = relative accuracy (e.g. 0.10 for ±10 %)

General: $n = z^2 p(1-p)/L^2$; Error limit $L = z\sqrt{p(1-p)/n}$

Procedure for conducting a work-sampling study

1. Define the objective & activities to be recorded (working, idle, delay-A, delay-B …); decide the confidence level and accuracy.
2. Make a pilot study (50–100 observations) to get a preliminary estimate of p; then compute n.
3. Decide number of observers, days and observations per round → n = (observations per round) × (rounds per day) × (days).
4. Fix **random observation times** using a random-number table / computer (so that observations do not synchronise with the work pattern).
5. Design the observation sheet, brief the workers; make the tours at the random times and record the state at the instant of sight.
6. Compute p = (number of observations in the activity / total observations); check the accuracy with control limits $p \pm 2\sigma$; if outside → take more observations.
7. Report: % utilisation, % delay, allowances; standard time if required.

Standard time from work sampling

Total time observed = T (min); Working % = P; Units produced = N; Rating = R %; Allowance = A %.

Normal time / unit = (T × P × R) / N; Standard time / unit = Normal time × (1 + A)

Problems

P1. A pilot study of 100 observations on a machine shows it idle in 20 observations. Find the number of observations for 95 % confidence and (a) ±5 % absolute accuracy; (b) ±10 % relative accuracy.

Sol: p = 0.20. (a) $n = 4 \times 0.2 \times 0.8 / (0.05)^2 = 0.64 / 0.0025 =$ **256 observations.** (b) $n = 4 \times (1 - 0.2) / (0.10^2 \times 0.2) = 3.2 / 0.002 =$ **1600 observations.**

P2. In a study of 400 observations, a machine was found working in 320 and idle in 80. Find the utilisation and the accuracy at 95 % confidence.

Sol: p(working) = 320/400 = 0.80. $\sigma = \sqrt{(0.8 \times 0.2 / 400)} = 0.02$; 95 % limit = $\pm 2\sigma = \pm 0.04$ → utilisation = **80 % ± 4 %** (76 %–84 %).

P3. Work-sampling data for an operator (8 h day, 480 min): 400 random observations → 80 idle. In the day 120 units were produced. Rating 110 %, allowance 15 %. Find the standard time per unit.

Sol: working % = 320/400 = 80 %. Working time = 480 × 0.80 = 384 min. Normal time = 384 × 1.10 / 120 = **3.52 min.** Standard time = 3.52 × 1.15 = **4.05 min / unit.**

P4. A study of 6 machines is to be made for 5 days with a 95 % confidence, ±5 % accuracy, assuming p = 0.25. How many rounds per day?

Sol: $n = 4 \times 0.25 \times 0.75 / 0.0025 = 300$ observations. Each round = 6 observations → rounds = 300 / 6 = 50 → per day = 50 / 5 = **10 rounds / day.**

8. Quick Revision & Probable Questions

Memory hooks

- Method study: SELECT-RECORD-EXAMINE-DEVELOP-DEFINE-INSTALL-MAINTAIN; ECRS; 5W+1H questioning.
- Symbols: O (operation) T (transport) I (inspection) D (delay) S (storage).
- Barnes: Body – Work place – Tools. Classes of movement: 1 finger … 5 body.
- Wink = 1/2000 min; Cyclegraph = path; Chronocyclegraph = path + speed.
- Work sampling: $n = 4p(1-p)/L^2$ (95 %).

1. Define method study. Explain its objectives and the steps in the procedure.
2. Draw the symbols used in process charts. Differentiate the outline process chart and the flow process chart.
3. Explain flow diagram, string diagram and travel chart with sketches.
4. What is a multiple activity chart? How many machines can one operator attend?
5. State the principles of motion economy under the three groups. Classify the movements.
6. What are therbligs? Explain micromotion study and the SIMO chart.
7. Explain cyclegraph and chronocyclegraph. How is the improved method defined and installed?
8. Define work measurement; list its techniques. What is work sampling? Derive the sample size formula and solve numericals.